SMART HOME AUTOMATION SYSTEM

An IoT-Based Edge Security Solution with ESP32 & Blynk Cloud Integration

Coursework Mini-Project Report

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Field: Internet of Things (IoT) & Embedded Systems
Platform Environment: Wokwi Smart System Simulator
Cloud Middleware: Blynk IoT Ecosystem

1. Executive Summary & Introduction

Modern residential frameworks increasingly rely on automation and smart ecosystems to maintain structural integrity, physical defense, and ambient environmental standards. This project details the deployment of an automated, real-time Internet of Things (IoT) Home Security System utilizing the high-performance ESP32 microcontroller, compiled and evaluated within the Wokwi hardware emulation suite, and managed seamlessly via the Blynk IoT Cloud Infrastructure.

The engineered architecture offers round-the-clock telemetry acquisition capturing environmental shifts (temperature and humidity) alongside instant defensive perimeter event capture (combustible gas leaks, open fire occurrences, and unexpected motion breaches). Through dynamic edge-computing constraints programmed onto the microchip, the architecture processes sensor data instantly to toggle defensive peripherals locally (e.g., triggering alarm buzzers or opening exhaust relay mechanisms) while simultaneously passing telemetry packets over standard wireless communication protocols to a secure mobile/web user dashboard.

2. System Architecture & Components

The foundational hardware layer combines precise digital and analog modules mapped into the ESP32 processing hub. Below is an analytical review of the component list selected for physical orchestration:

Component Name	Technical Purpose & Operational Specification
ESP32 Microcontroller	Acts as the principal processing core. Equipped with integrated Wi-Fi and Bluetooth stacks, it hosts the local edge logic, queries sensors, executes interrupt cycles, and maintains standard MQTT connections to the Blynk platform.
PIR Motion Sensor	Performs spatial infrared monitoring. Toggles a digital output high upon detection of dynamic heat signatures, identifying active perimeter breaches or unauthorized room occupancy.
Flame Sensor	Utilizes an infrared receiving photodiode designed to capture wave-lengths associated with standard ignition sources. Operates on a rapid active-LOW binary response mechanism for immediate fire awareness.
DHT22 Sensor	Provides low-power digital measurement of room thermodynamics, outputting precise calibrated temperature and relative humidity metrics via a proprietary single-wire serial stream.
MQ-2 Gas Sensor	An analog electrochemical device sensitive to combustible elements including

	LPG, smoke, butane, and propane. Operates across a wide ADC measurement spectrum for granular contamination assessment.
Active Buzzer	A structural acoustic transducer that serves as a localized audible warning beacon. Programmed to pulsate at varying frequencies depending on threat severity levels.
Relay Module	An electronic isolating switch configured as active-HIGH. Provides a safe interface to activate auxiliary physical equipment such as localized exhaust blowers or chemical suppression units.

3. Wiring Topography & Pin Mappings

The operational layout demands dedicated General Purpose Input/Output (GPIO) mapping to ensure data fidelity and avoid multiplexing contentions. A critical engineering constraint requires isolating analog ingestion to ADC1 lines, as ADC2 becomes inaccessible when Wi-Fi communication stacks are active on the ESP32 chip.

> *CRITICAL WARNING: GPIO 34 is hardwired internally as an input-only line lacking hardware pull-up capability. It is specifically reserved for incoming analog telemetry from the MQ-2 module. Attempting to program GPIO 34 as a digital output will result in firmware instantiation failure.*

Peripheral Module	ESP32 Pin Mapping	Electrical Parameter / Operational Notes
DHT22 (Data Line)	GPIO 27	Requires external 10kΩ pull-up resistor to 3.3V bus to prevent floating logic.
PIR Sensor (Digital Out)	GPIO 15	Configured as standard digital input; logic HIGH flags an active breach.
Flame Sensor (Digital Out)	GPIO 4	Active-LOW response; a value of 0 flags a confirmed fire outbreak.
MQ-2 Gas Sensor (Analog Out)	GPIO 34	Mapped to internal Successive Approximation Register (SAR) ADC1 channel.
Relay Module (Input Line)	GPIO 2	Configured active-HIGH; shifts relay contacts to

		Normally Open (NO).
Active Buzzer (Positive Terminal)	GPIO 14	Digital output mode; driven directly by logic states for frequency alarms.
System VCC Rails	3.3V / 5.0V	Sensors balanced across power rails corresponding to data sheet parameters.
System Ground Plane	GND Nodes	Common ground system across all independent sensor breakout boards.

4. Cloud Infrastructure Configuration (Blynk IoT)

The remote management component of the system relies on the Blynk IoT platform. Connecting edge devices with cloud assets requires structured provisioning of Virtual Pins to transmit multi-variable data over a single connection interface.

4.1 Datastream Matrix Design

Virtual pins decouple hardware abstraction layers from real-time data visualizers. The template layout is engineered using the following constraints:

Telemetry Element	Virtual Pin	Data Type	Min Range	Max Range
Gas Leak Level	V1	Integer	0	1
Flame Detection Flag	V3	Integer	0	1
Intruder Motion Status	V4	Integer	0	1
Thermodynamic Temperature	V5	Double / Float	-40°C	80°C
Relative Ambient Humidity	V6	Double / Float	0%	100%
Auxiliary Relay State	V7	Integer	0	1

4.2 Interface & Event Handlers

Web and Mobile Layouts: 1. Dashboard Assembly:

Visual indicators (LED widgets) link to binary nodes V1, V3, V4, and V7 to provide high-visibility operational updates. Dynamic metrics for V5 and V6 use arc gauges to show live thermal and humidity fluctuations.

Push Notification Protocols: 2. Asynchronous Alert Logic:

An active event handler named `warning_message` is built directly into the system template. This allows the microchip to run critical cloud actions using the command sequence `Blynk.logEvent("warning_message", "...")` when sensor states pass configured safety boundaries.

5. Edge Computing Logic & Safety Alarms

The local firmware script loop processes inputs against strict safety thresholds, triggering immediate local defense measures and remote notifications without relying on cloud-side evaluation loops.

5.1 Fire Outbreak (Flame Vector)

The system continuously polls GPIO 4. Upon discovering an active-LOW state (`digitalRead(4) == LOW`), indicating an external fire source, the system immediately launches the following emergency sequence:

- Forces the localized automation relay (GPIO 2) to HIGH to trigger fire suppression or ventilation systems.
- Generates an audible warning through the piezo-buzzer (GPIO 14) with an alternating 550Hz acoustic frequency signal.
- Transmits a priority warning flag to the Blynk event engine, delivering a 'Fire Hazard Detected' alert directly to user mobile devices.

5.2 Combustible Gas Leak (MQ-2 Vector)

The analog interface reads raw voltage levels from the environment using 12-bit depth resolution (spanning values from 0 to 4095) on GPIO 34. The firmware normalizes this scale down to an 8-bit space (0 to 255).

If the filtered output climbs beyond a safety limit of 165, the system responds immediately: it flips the exhaust fan relay, drives the active buzzer at a distinct 1000Hz frequency to distinguish it from a fire alarm, and pushes a 'Gas Leakage Warning' alert out to the cloud dashboard.

5.3 Thermal Critical Limits (DHT22 Vector)

When ambient temperatures climb to or exceed 50°C, the system triggers safety protocols to protect property. It engages the cooling relay module to run ventilation fans and logs a temperature event on the network. The alert condition remains locked until temperature readouts drop safely back below the 50°C benchmark.

5.4 Perimeter Intrusion (PIR Vector)

The digital state machine tracks physical security. When the PIR module registers movement (`digitalRead(15) == HIGH`), the system flags a possible intruder breach. The firmware uses an internal tracking state variable to prevent sending duplicate notifications, ensuring the system issues a single clear alert without overloading the user's device.

6. Implementation, Testing & Troubleshooting

6.1 Virtual Environment Execution

Validation testing was performed using the Wokwi development platform. Network connection routines were established by binding the system to the standard virtual wireless access point `Wokwi-GUEST` using open-security authentication parameters. The firmware successfully handles full sensor loops, runs edge alarm triggers, and maintains stable data transfers with Blynk servers.

6.2 Systematic Diagnostics Guide

Observed System Error	Root-Cause Diagnostics & Resolution Path
Wireless Network Dropping / Timeout	Ensure the simulation configuration file utilizes the precise 'Wokwi-GUEST' SSID parameters and leave the security passphrase empty.
Cloud Dashboard Disconnected State	Verify that the `BLYNK_TEMPLATE_ID`, `BLYNK_TEMPLATE_NAME`, and `BLYNK_AUTH_TOKEN` fields match the credentials generated in the Blynk developer console.
Gas Leak Alert Latency or False Alarms	Adjust the software threshold variable (default 165) in the setup code to compensate for simulation variances or balance specific baseline air quality environments.
DHT22 Reporting Invalid 'NaN' Streams	Inspect data line traces on GPIO 27. Ensure the model includes a 10kΩ pull-up resistor to the 3.3V rail to prevent floating signal states.
Push Alerts Failing to Deliver	Confirm that the event configuration in the Blynk console includes an active template string named exactly 'warning_message' with notifications enabled.

7. Conclusion

This mini-project demonstrates a highly responsive, cost-effective, and dependable approach to modern residential security. By combining local edge processing on the ESP32 with cloud-based monitoring via Blynk IoT, the architecture ensures critical response mechanisms operate instantly, keeping homeowners informed no matter where they are. Testing within the Wokwi simulation environment confirms the system's viability for real-world deployment, providing a solid foundation for advanced smart home safety integrations.